

IIT JAM 2011 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

15

TOTAL MARKS

90 M

TEST DURATION

60 Min

PAPER PATTERN

MCQ

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2011 • Q1 • ID: JAM_2011_Q1)

MCQ

+6 M

Neg: -2.0

Q.1 Let $a_n = \sum_{k=1}^n \frac{n}{n^2+k}$, for $n \in \mathbb{N}$. Then the sequence $\{a_n\}$ is

(A) Convergent

(B) Bounded but not convergent

(C) Diverges to ∞ (D) Neither bounded nor diverges to ∞

Question 2 (JAM 2011 • Q2 • ID: JAM_2011_Q2)

MCQ

+6 M

Neg: -2.0

Q.2 The number of real roots of the equation $x^3 + x - 1 = 0$ is

(A) 0

(B) 1

(C) 2

(D) 3

Question 3 (JAM 2011 • Q3 • ID: JAM_2011_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 The value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n^2+kn}}$ is

(A) $2(\sqrt{2}-1)$ (B) $2\sqrt{2}-1$ (C) $2-\sqrt{2}$ (D) $\frac{1}{2}(\sqrt{2}-1)$

Question 4 (JAM 2011 • Q4 • ID: JAM_2011_Q4)

MCQ

+6 M

Neg: -2.0

Q.4 Let V be the region bounded by the planes $x=0$, $x=2$, $y=0$, $z=0$ and $y+z=1$. Then the value of the integral $\iiint_V y \, dx \, dy \, dz$ is

(A) $\frac{1}{2}$ (B) $\frac{4}{3}$

(C) 1

(D) $\frac{1}{3}$

Question 5 (JAM 2011 • Q5 • ID: JAM_2011_Q5)

MCQ

+6 M

Neg: -2.0

- Q.5 The solution $y(x)$ of the differential equation $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$ satisfying the conditions $y(0)=4, \frac{dy}{dx}(0)=8$ is
- (A) $4e^{2x}$ (B) $(16x+4)e^{-2x}$ (C) $4e^{-2x}+16x$ (D) $4e^{-2x}+16xe^{2x}$

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Question 6 (JAM 2011 • Q6 • ID: JAM_2011_Q6)

MCQ

+6 M

Neg: -2.0

- Q.6 If y^a is an integrating factor of the differential equation $2xy dx - (3x^2 - y^2) dy = 0$, then the value of a is
- (A) -4 (B) 4 (C) -1 (D) 1

Question 7 (JAM 2011 • Q7 • ID: JAM_2011_Q7)

MCQ

+6 M

Neg: -2.0

- Q.7 Let $\vec{F} = ay\hat{i} + z\hat{j} + x\hat{k}$ and C be the positively oriented closed curve given by $x^2 + y^2 = 1, z = 0$. If $\oint_C \vec{F} \cdot d\vec{r} = \pi$, then the value of a is
- (A) -1 (B) 0 (C) $\frac{1}{2}$ (D) 1

Question 8 (JAM 2011 • Q8 • ID: JAM_2011_Q8)

MCQ

+6 M

Neg: -2.0

- Q.8 Consider the vector field $\vec{F} = (ax + y + a)\hat{i} + \hat{j} - (x + y)\hat{k}$, where a is a constant. If $\vec{F} \cdot \text{curl} \vec{F} = 0$, then the value of a is
- (A) -1 (B) 0 (C) 1 (D) $\frac{3}{2}$

Question 9 (JAM 2011 • Q9 • ID: JAM_2011_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 Let G denote the group of all 2×2 invertible matrices with entries from $\mathbb{R}$. Let

$$H_1 = \{A \in G : \det(A) = 1\} \text{ and } H_2 = \{A \in G : A \text{ is upper triangular}\}.$$

Consider the following statements:

P : H_1 is a normal subgroup of G

Q : H_2 is a normal subgroup of G .

Then

(A) Both P and Q are true

(B) P is true and Q is false

(C) P is false and Q is true

(D) Both P and Q are false

Question 10 (JAM 2011 • Q10 • ID: JAM_2011_Q10)

MCQ

+6 M

Neg: -2.0

Q.10 For $n \in \mathbb{N}$, let $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$. Then the number of units of $\mathbb{Z}/11\mathbb{Z}$ and $\mathbb{Z}/12\mathbb{Z}$, respectively, are

(A) 11, 12

(B) 10, 11

(C) 10, 4

(D) 10, 8

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Question 11 (JAM 2011 • Q11 • ID: JAM_2011_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let A be a 3×3 matrix with $\text{trace}(A) = 3$ and $\det(A) = 2$. If 1 is an eigenvalue of A , then the eigenvalues of the matrix $A^2 - 2I$ are

(A) $1, 2(i-1), -2(i+1)$

(B) $-1, 2(i-1), 2(i+1)$

(C) $1, 2(i+1), -2(i+1)$

(D) $-1, 2(i-1), -2(i+1)$

Question 12 (JAM 2011 • Q12 • ID: JAM_2011_Q12)

MCQ

+6 M

Neg: -2.0

Q.12 Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation, where $n \geq 2$. For $k \leq n$, let

$$E = \{v_1, v_2, \dots, v_k\} \subseteq \mathbb{R}^n \text{ and } F = \{Tv_1, Tv_2, \dots, Tv_k\}.$$

Then

- (A) If E is linearly independent, then F is linearly independent
- (B) If F is linearly independent, then E is linearly independent
- (C) If E is linearly independent, then F is linearly dependent
- (D) If F is linearly independent, then E is linearly dependent

Question 13 (JAM 2011 • Q13 • ID: JAM_2011_Q13)

MCQ

+6 M

Neg: -2.0

Q.13 For $n \neq m$, let $T_1: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T_2: \mathbb{R}^m \rightarrow \mathbb{R}^n$ be linear transformations such that $T_1 T_2$ is bijective. Then

- (A) $\text{rank}(T_1) = n$ and $\text{rank}(T_2) = m$
- (B) $\text{rank}(T_1) = m$ and $\text{rank}(T_2) = n$
- (C) $\text{rank}(T_1) = n$ and $\text{rank}(T_2) = n$
- (D) $\text{rank}(T_1) = m$ and $\text{rank}(T_2) = m$

Question 14 (JAM 2011 • Q14 • ID: JAM_2011_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 The set of all x at which the power series $\sum_{n=1}^{\infty} \frac{n}{(2n+1)^2} (x-2)^{3n}$ converges is

- (A) $[-1, 1)$
- (B) $[-1, 1]$
- (C) $[1, 3)$
- (D) $[1, 3]$

Question 15 (JAM 2011 • Q15 • ID: JAM_2011_Q15)

MCQ

+6 M

Neg: -2.0

Q.15 Consider the following subsets of $\mathbb{R}$:

$$E = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}, \quad F = \left\{ \frac{1}{1-x} : 0 \leq x < 1 \right\}.$$

Then

- (A) Both E and F are closed
(B) E is closed and F is NOT closed
(C) E is NOT closed and F is closed
(D) Neither E nor F is closed

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OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2011 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	A
2	MCQ	+6	B
3	MCQ	+6	A
4	MCQ	+6	D
5	MCQ	+6	B
6	MCQ	+6	A
7	MCQ	+6	A
8	MCQ	+6	D

Q. No.	Type	Marks	Official Key
9	MCQ	+6	B
10	MCQ	+6	C
11	MCQ	+6	D
12	MCQ	+6	B
13	MCQ	+6	D
14	MCQ	+6	C
15	MCQ	+6	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.